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TL;DR: *E-commerce search is broken because it speaks "database" while customers speak "human." By applying Marketing Myopia theory, I built a hybrid PoC that uses LLM Intent Translation and Vector Embeddings to turn vague queries into precise, deterministic results. The result? No more "Zero Result" pages and a search that actually understands the user.*

Semantic Search & Intent Translation: The Future of E-Commerce

During my tenure at SSE, I attended a workshop with a company whose name escapes me, but one takeaway stuck: E-commerce search engines are failing.

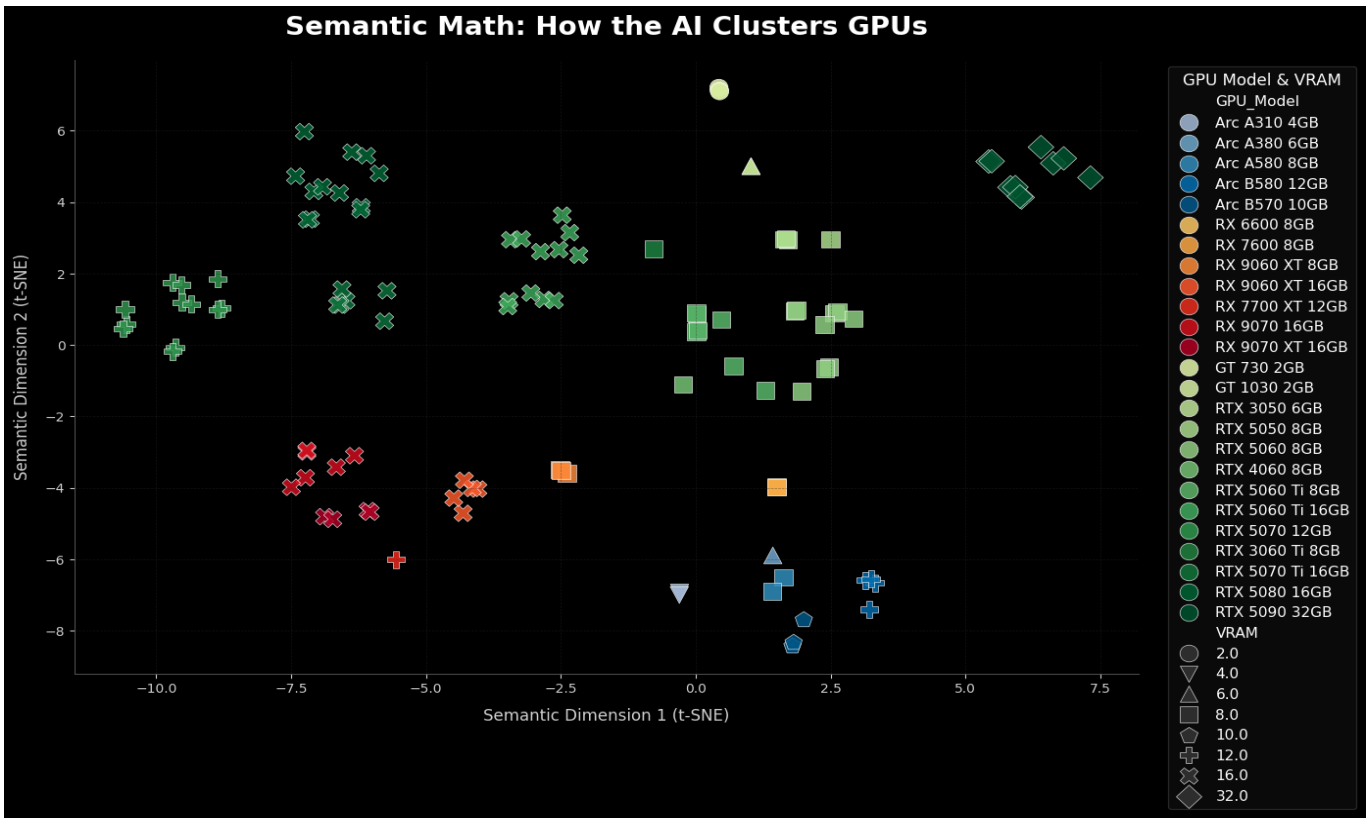
What do I mean by that? Most e-commerce sites today still rely on rigid keyword matching. It's binary. You must type the exact string of text that matches the database, or you get "No results found."

Think about why this matters. Have you ever tried to find a product you can't quite name? Or worse, tried to buy a gift for a loved one in a niche you know nothing about? You end up guessing, buying something "off-trend," and having to watch their polite, fake smile while hoping they see you "put your heart into it."

Semantic Search & Intent Translation is the fix for that "fake smile" moment.

To solve this, we have to change how we treat data. Here's the brief version of how it works:

1. **Vectorization**: We take a dataset of every SKU and pass it through a text embedder (I used BAAI/BGE-M3). A massive benefit of this specific model is multi-linguality, meaning that users can use their native language for search and get perfectly matched results.
2. **The Map**: This converts text, which is notoriously messy to analyze, into a coordinate (a vector). Imagine plotting every product on a massive multi-dimensional graph.
3. **The Result**: The model now "knows" the relationship between products based on their characteristics, not just their names.



But as the saying goes: *Garbage in, garbage out*. You need high-quality data for this to work!

Why an LLM is integral for this

In theory, a vectorized database is enough. In reality, customers are vague, they make typos, or they describe "vibes" rather than specs. This is where LLMs solve the problem perfectly. We use the LLM to parse the query using a highly detailed prompt.

I've used prompts like this to bridge the gap:

> **"You are an expert GPU hardware search assistant operating in 2026. Your goal is to-translate a casual user query into a dense, high-signal technical search string..."**

We can't just use any model, we must balance speed and reasoning quality. And since LLMs can occasionally "hallucinate," we backstop them with deterministic filtering. If a customer mentions "16 GB," we extract that as a hard requirement. Even if the LLM gets creative, the results stay grounded in reality.

The "Marketing Myopia" Moment

In his classic paper Marketing Myopia, Theodore Levitt argued that industries stagnate when they define themselves by their products (e.g., "we sell train rides") rather than the customer need ("we provide transportation").

And in E-commerce, I believe many firms are currently facing a similar Myopia. Many platforms believe they are in the "Keyword Matching" business. They aren't. They are in the "Intent Fulfillment" business. By focusing on the utility of the search bar rather than the psychology of the seeker, we create a "one-size-fits-none" experience.

The Two Types of Intent:
The "Myopia" happens when we treat every user as a data entry clerk. In reality, users fall into two distinct psychological groups:

1. **The Specialist Shopper**: Her requirement is Precision. She has the technical vocabulary to navigate filters and expects a high-fidelity tool. To her, search is a scalpel - and most importantly: FUN!
2. **The Generalist**: Her requirement is Interpretation. She lacks the domain-specific language and expects the system to understand her context. To her, search should be a curator.

The solution is not to provide two search boxes. Or to choose between one or the other. It's to implement a hybrid approach. Build a system that integrates the benefits of the semantic search engine alongside traditional keyword search!

Summary of The Benefits

1. **Hybrid search**: Satisfies all customer intents across any written language.
2. **Reduce Search Abandonment**: Eliminate the "Zero Results" dead-end.
3. **Optimize Inventory**: Use weighted semantic math to prioritize high-margin items or overstock that fits the user's actual needs.
4. **Unlock Data-Driven Insights**: Analyze what users are asking for in plain language text to inform future inventory purchasing.

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The Why Question

So, why isn't everyone doing this? Some are. Zalando is a great example, they've built a robust engine that combines the best of both worlds. But for most retailers, it's a hurdle of complexity and computational cost. But here is my challenge to business leaders: Don't ignore this because it feels out of reach.

Now, I'm not underestimating the technical friction here. I know that building a working PoC in a couple of days using BGE-M3 and Ollama is worlds apart from launching a globally scalable, low-latency engine on a live e-commerce site. That transition is a massive, capital-intensive engineering challenge.

But here is my point to business leaders: The barrier to entry isn't algorithmic, it's the foundation. The real reason these projects stall is usually organizational. It requires the cross-functional will to overhaul legacy data pipelines, sanitize messy vendor metadata, and justify the long-term cloud compute costs.

I've felt this tension firsthand. My early ambition to vectorize 'expert reviews' for every SKU proved that technical vision quickly hits a wall! If we want this to be sustainable, suppliers and retailers have to integrate their data pipelines vertically. To summarize, I believe that the engineers can build the engine, but the business must first commit to the structural shift and budget required to fuel this transition, or risk falling behind.

Defining KPIs to Measure Success

These KPIs could be used when evaluating the before-and-after, determining success or failure of the implementation:

General KPIs

1. Search Abandonment: The implementation should reduce the amount of users leaving the site after a search.
2. Search to Cart Ratio: Do A/B testing on the conversion of users using the hybrid search vs the keyword search. Theoretically, we should see higher conversion rates for shoppers who are ready to buy but lack domain knowledge.
3. AOV: The average order value should increase. We know intent, thus we can programmatically return bundles or high margin items if it matches intent.

Operation KPIs

1. Reduction in manual labour: Long term, when data pipelines are automated from the supplier and the team no longer needs manually update datasets, manual labour should decrease. E-commerce teams today may manually create keyword mappings, e.g. cheap = clearance. But vector mapping automates this. Hence, calculate the time saved.
2. Customer Support: Collect data on the amount of questions pre semantic search and post. The implementation should reduce the load.

Inventory and Data Insights KPIs

1. Return Rates: Semantic search matches intent, hence it should reduce return rates.
2. Demand Identification: Semantic search allows for analysis of intent clusters from the LLM logs. We can use that data as a procurement signal, to either acquire missing assortment or

help suppliers develop the right products. A keyword search would have simply logged the search as a fail if it returned no results.

Strategic Recommendation

1. **Product Information Management (PIM)**: If the vector is built on poor quality metadata, the results will be as bad or worse than legacy search. Therefore, clean and reliable metadata is top priority. There must be strict data protocols. Small firms with low bargaining power will encounter difficulty in demanding vendors to comply if data pipelines have not been established already.
2. **Hybrid Search Engine**: Not every search can be routed through an LLM, that would be too costly. Instead, the intent translation should only be used for vague or natural language queries.
3. **Keep Track of The Big 3**: The whole point of the semantic search engine is to fix a broken conversion funnel. Thus, every single query should lead to an appropriate result for the customer. Every query that is written in natural language and leads to conversion or product exploration is a win, measure this! The added benefit of translating intent with business logic should cause an increase in AOV, so measure this closely.

PoC Results & Comparison

Example of search abandonment

Legacy search engines fail when users submit high-intent, natural language queries.

The Query: "I want a card that will last me the next 5 years for AAA gaming. I'm currently on an old 1080p monitor but plan to upgrade to 4K soon. What's the best investment right now?"

The Retailer Response:

Business Impact: Friction. The unmotivated customer gets frustrated and leaves.

The solution? AI Intent Translation

My PoC uses Vector Embeddings and LLM Query Expansion to translate that same vague request into deterministic hardware requirements (e.g., VRAM >= 12GB, High-end Ray Tracing).

- **Scenario 1**: The High-Performance Path

The system identifies the "4K upgrade" intent and prioritizes high-VRAM, future-proofed cards.

Ask for a GPU naturally:

I want a card that will last me the next 5 years for AAA gaming. I'm currently on an old 1080p monitor but plan to upgrade to 4K soon. What's the best investment right now?

AI Intent Translation: `Future-proof Enthusiast GPU, 4K-capable, 1080p/4K upgrade path, 5-year longevity, high-end ray tracing, DLSS/FSR support, mixed workloads (gaming + productivity), 12GB+ VRAM baseline, Blackwell/Ada/RDNA 4/Alchemist architecture, 16GB+ VRAM premium tier`

Hard Filter Enforced: `>= 12GB VRAM`

Top Recommendations based on Semantic Math:

Name	↑ Current Price (SEK)	VRAM (GB)
Gigabyte GeForce RTX 5070 Ti WINDFORCE OC SFF 16GB	11490 SEK	16
ASUS PRIME GeForce RTX 5070 Ti 16GB OC	11790 SEK	16
ASUS ROG Strix GeForce RTX 5070 Ti 16GB OC	11990 SEK	16
ASUS ROG Strix GeForce RTX 5070 Ti 16GB	12990 SEK	16
Gigabyte Aorus GeForce RTX 5070 Ti MASTER 16GB	13199 SEK	16

- **Scenario 2: Dynamic Budget Sensitivity**

When we append "I don't want to spend too much!" to the query, the AI re-evaluates. It maintains the technical requirements (16GB VRAM) but filters for high-value/mid-range alternatives.

Ask for a GPU naturally:

I want a card that will last me the next 5 years for AAA gaming. I'm currently on an old 1080p monitor but plan to upgrade to 4K soon. What's the best investment right now? I dont want to spend too much!

AI Intent Translation: `Mid-to-High-end Enthusiast GPU, 1080p/4K future-proofing, 5-year longevity, balanced performance, 1440p+ capable, ray tracing support, DLSS/FSR 3 compatibility, 16GB+ VRAM minimum, Ada Lovelace/Blackwell or RDNA 4 architecture, budget-conscious premium`

Hard Filter Enforced: `>= 16GB VRAM`

Top Recommendations based on Semantic Math:

Name	Current Price (SEK)	VRAM (GB)
ASUS TUF Gaming Radeon RX 9060 XT OC Edition 16GB	6649 SEK	16
ASRock Radeon RX 9070 16GB Steel Legend OC	8999 SEK	16
ASRock Radeon RX 9070 Challenger 16GB	8490 SEK	16
ASRock Radeon RX 9060 XT Challenger 16GB	4590 SEK	16
ASUS Dual Radeon RX 9060 XT 16GB	4690 SEK	16

Disclaimers & Future Roadmap: Scaling the PoC

The PoC currently utilizes local inference and generalized datasets, which introduces latency and data homogenization. And while the current vector search effectively identifies broad product categories, it requires high-fidelity, SKU-specific metadata and a dedicated scoring layer to achieve production-grade accuracy.

The remaining constraints could be resolved by:

1. **Hybrid Routing via Query Intent Classifier**: To optimize latency and cost, the system should utilize a classifier to "route" queries. Vague or natural language inputs are sent for semantic LLM processing, while structured or SKU-specific queries bypass the LLM for direct, low-latency database retrieval.
2. **Precision Scoring**: We replace basic string matching with deterministic Function Calling and implement a weighted re-ranking formula. This allows the system to balance semantic relevance with real time business logic. For example, prioritizing specific price targets or stock availability over technical specification matches.

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